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New Cretaceous and Early Tertiary paleomagnetic results from Xining-Lanzhou basin, Kunlun and Qiangtang blocks, China: Implications on the geodynamic evolution of Asia

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Abstract. We present the results of a paleomagnetic study of 360 cores, drilled at three different areas from the Tibetan Plateau: 13 sites from Lower Cretaceous red beds in the Xining-Lanzhou basin, around Xining and Lanzhou cities, southwest of the Qilian mountains (36.2°N, 103.5°E); 13 sites from Cretaceous red beds in the Kunlun block, near Maqin (34.5°N, 100.1°); and 9 sites from Paleocene/lower Eocene red beds in the Qiangtang block, near Fenghuoshan (34.5°N, 92.8°E). Thermal demagnetization of the samples allowed us to isolate a high-temperature component which passes both positive reversal and fold tests for all formations. The corresponding paleopoles lie at 50.3°N, 195.5°E (A_{95}=4.6°) for Xining-Lanzhou, 80.1°N, 281.2°E (dp=7.8°, dm=12.7°) for Maqin, and 62.6°N, 210.5°E (dp=3.9°, dm=6.8°) for Fenghuoshan. We discuss these in the frame of a new paleogeographic reconstruction of the Cretaceous paleoposition of the blocks forming the Asian mosaic. We conclude that the Xining-Lanzhou area could not be part of the North China Block but rather was associated with the Tarim-Qaidam assemblage. Paleomagnetic data argue in favor of a Qaidam-Kunlun-Tarim-Junggar assemblage in the Cretaceous, significantly to the south of its current position with respect to the Asian continent (Siberia, Mongolia and North China). The large N-S convergence (800±500 km) implied since the Cretaceous appears to be far larger than could be absorbed in the Altay range to the north and Qilian Shan to the east (of the order of 300 km). Part of this motion could have occurred along a large left-lateral strike slip fault system, which may connect with the Mongol-Okhotsk suture to the northeast. Verifying this hypothesis will require new geologic and paleomagnetic data from these remote regions.

1. Introduction

Several paleomagnetic studies carried out in China during the last decade have improved our knowledge of the accretion history of the various Asian blocks [e.g., see *Enkin et al.*, 1992) and of the tectonic modifications of this assemblage under the effect of the collision of India [*Molnar and Tapponnier*, 1975; *Tapponnier et al.*, 1986; *Chen et al.*, 1993b; *Yang and Besse*, 1993].

Regarding the central Asian blocks (Figure 1), recent paleomagnetic studies of Mesozoic formations in Tibet [*Pozzi et al.*, 1982; *Westphal et al.*, 1983; *Achache et al.*, 1984; *Lin and Watts*, 1988; *Otofuji et al.*, 1989; *Huang et al.*, 1992; *Chen et al.*, 1993a), the Tarim basin [*Li et al.*, 1988; *Chen et al.*, 1992], the Ordos basin [*Yang et al.*, 1991], the North China Block [*Pruner*, 1988; *Zhao et al.*, 1990; *Zheng et al.*, 1991; *Ma et al.*, 1993; *Yang et al.*, 1991], the South China Block [*Kent et al.*, 1987; *Zhu et al.*, 1988; *Enkin et al.*, 1991b], and the Junggar basin [*Chen et al.*, 1991] allowed *Chen et al.* [1993b] to propose a reconstruction of central Asia prior to the collision of India. More recent studies on the Turfan and Junggar basins [*Cogné et al.*, 1995] and the Tarim basin [*Gilder et al.*, 1996] have provided results that are in line with this model. The Cretaceous is the period for which the most numerous and best quality data are available. It has also the advantage that the Eurasian plate exhibits a stable apparent polar wander path (APWP), the so-called Cretaceous "hairpin turn" [*Besse and Courtillot*, 1991; *Enkin et al.*, 1991a], spanning the entire Cretaceous, when accurate dating of sampled formations, which are mostly continental red sandstones, is often wanting. *Chen et al.* [1993b] proposed that the total amount of convergence between India and Siberia since the time of collision of India has been largely accommodated within the mountain ranges (Himalaya, Kunlun, Tian Shan, Altay, etc.) which separate the different blocks (Figure 1). This interpretation, of course, depends on determination of the relative paleopositions of the different blocks for different epochs prior to the collision and on the number of paleomagnetic pole determinations within each block.

In this respect, the movement of the Qaidam, Kunlun, and Qiangtang Blocks are rather poorly constrained from a paleomagnetic point of view, with the movement of the Qaidam Block being defined by a single preliminary result by *Frost et al.* [1995], based on 21 specimens from two sites. For the Qiangtang Block, *Lin and Watts* [1988] have studied samples from the Paleocene (?) - lower Eocene Fenghuoshan Group red beds, the Carboniferous Dagangou Formation and

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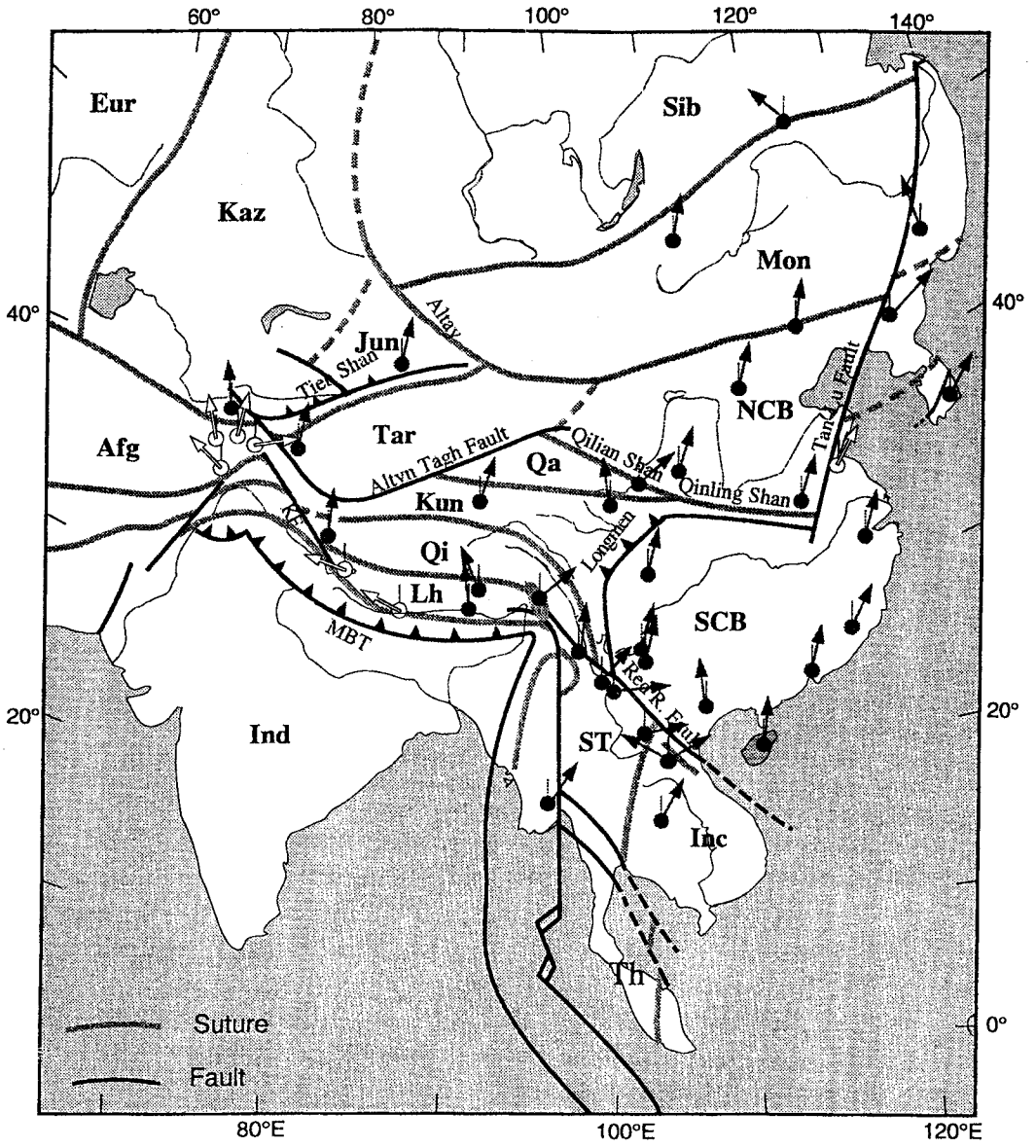


Figure 1. Simplified map of southeast Asia showing the main sutures and faults (KF, Karakorum fault; MBT, Main Boundary Thrust). The major blocks are: Afg, Afganistan; Eur, Eurasia; Inc, Indochina; Ind, India; Jun, Junggar; Kaz, Kazakhtan; Kun, Kunlun; Lh, Lhasa; Mon, Mongolia; NCB, North China Block; Qa, Qaidam; Qi, Qiangtang; Sib, Siberia; SCB, South China Block; ST, Shan Tai; and Tar, Tarim; Th, Thailand. Also indicated are the main locations where Cretaceous paleomagnetic data are available (Table 5). Declination is indicated by arrows (a thin line indicates geographic north, it is vertical because of the projection selected). Open arrows are for areas suspected of major local scale deformation or unreliable paleomagnetic data. Solid arrows are thought to be representative of the individual blocks at a larger scale.

Triassic dikes. For the Fenghuoshan locality, they obtained a characteristic remanent magnetization (ChRM) with reverse polarity only and no fold test. For the Kunlun block, no Mesozoic or lower Tertiary paleomagnetic studies have yet been performed.

In order to complement these preliminary studies, we report new paleomagnetic results from Lower Cretaceous to early Tertiary red continental sediments collected from three areas during field trips in 1993 and 1994 as part of a Sino-French cooperation: (1) around Xining and Lanzhou towns, in the Hekou group of the Xining-Lanzhou basin; (2) near Maqin town in the Kunlun block, somewhat south of the major Kunlun Fault and (3) in the Fenghuoshan Ranges along the road from Golmud to Lhasa, in the Fenghuoshan-Erdaogou Group of the Qiangtang block (Figure 2).

2. Geological Setting and Field Sampling

2.1. Qaidam Block: Xining-Lanzhou Section (Hekou Group, K_{1hk})

The Hekou group is composed of dark red sandstones interbedded with mudstones. The formation varies in thickness from a few hundred to about 2000 m and is widely exposed in

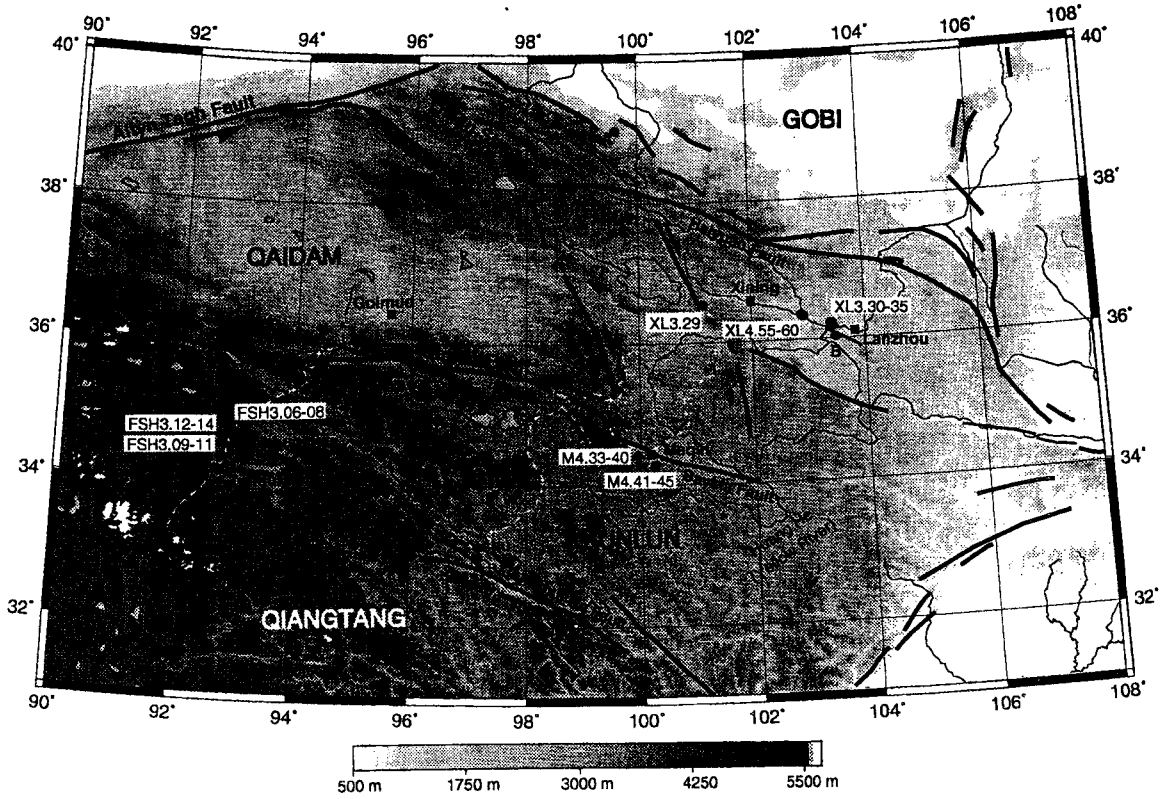


Figure 2. Simplified topographic map with major faults. Squares indicate town locations; circles are the paleomagnetic sampling sites with XL, sites of Xining-Lanzhou basin; A and B, sites of *Frost et al.* [1995]; M, sites of Maqin area; FSH, sites of Fenghuoshan-Erdaogou basin.

Qinghai and Gansu provinces. Following the Table of Chinese Stratigraphy [*Bureau of Geological and Mineral Resources of Qinghai Province (BGMRQP)*, 1991], the Qinghai-Xizang geological map [*Geological Institute of Chengdu (GIC)*, 1988], and the Qinghai geological map [*BGMRQP*, 1991], the age of this formation is determined mainly by pollen, ostracods, and charophyte fossils as Lower Cretaceous [*Yang et al.*, 1986; *Frost et al.*, 1995]. The Hekou group unconformably overlies Lower Paleozoic granites and is overlain by Upper Cretaceous red sandstones of the Minhe Group.

We sampled 13 sites (Figure 2): sites XL3.30 to XL3.35 at (36.2°N, 103.4°E) near Lanzhou, sites XL4.55 to XL4.60 at (36.3°N, 102.9°E), and site XL3.29 near Xining at (36.5°N, 101.1°E). Bedding attitudes vary from 30° to 50° toward the north and south (see Table 1), providing a favorable situation for application of the fold test. Our Xining-Lanzhou Basin results will be discussed together with those obtained by *Frost et al.* [1995] from two localities: one near Sunan in the Gansu Corridor (A, 39.0°N, 99.6°E) and the other near Lanzhou (B, 36.0°N, 103.5°E) (Figure 2).

Generally, 8 to 10 samples were taken at each site, using a gasoline-powered drill, and were oriented using solar and magnetic compass measurements in order to correct for the local magnetic field deviation. Both solar and magnetic readings were available for 21 cores, and the differences average -2.0°±1.6°. This is consistent with the value computed from the 1990 International Geomagnetic Reference Field (IGRF), corrected from secular variation, which is -1.7° in the Xining-Lanzhou area.

2.2. Kunlun Block: Maqin Section (Wanxiou Group, K_{1wn})

The Lower Cretaceous Wanxiou Group, which is well exposed in the Maqin area, is made up of a conglomerate member, more than 1500 m thick, overlain by red sandstones [*BGMRQP*, 1991]. The group unconformably overlies Permian limestones and dark red sandstones. The currently active left-lateral Baikla Fault (a branch of the major Kunlun Fault) crosses this area. According to the Qinghai geological map [*BGMRQP*, 1991], this group yields fossils of Lower Cretaceous age.

We sampled at two different localities: sites M4.33 to M4.40 to the west of Maqin town (34.5°N, 99.9°E) and sites M4.41 to M4.45 to the south of Maqin (34.4°N, 100.3°E). Bedding detail and site coordinates are listed in Table 1. Based on only five joint measurements of solar and magnetic azimuth, magnetic declination was found to average 1.3°±1.3°, in reasonable agreement with the value computed from the 1990 IGRF (-0.9°). Because of the small number of cores, we use the latter value rather than that determined from the cores.

2.3. Qiangtang Block: Fenghuoshan Section (Ter)

The Fenghuoshan group is exposed in the area of Tuotuohe and is composed of more than 3500 m of red sandstones, intercalated with mudstones, silstones, and thin lacustrine limestones, dated from charophytes and palynomorphs as

Table 1. Site Field Parameters

Site	N	Age	Locality	Geographic Coordinates	Bedding
			Xining-Lanzhou		
XL3.29	11	K_1	W.Xining	36°30'33"N, 101°08'52"E	318.1°/55N
XL3.30	8	K_1	N.Lanzhou	36°12'57"N, 103°25'29"W	335.8°/44N
XL3.31	8	K_1	N.Lanzhou	36°12'15"N, 103°26'13"W	92.0°/14S
XL3.32	8	K_1	N.Lanzhou	36°11'41"N, 103°26'32"W	38.8°/18S
XL3.33	6	K_1	N.Lanzhou	36°12'57"N, 103°25'29"W	156.9°/75S
XL3.34	8	K_1	N.Lanzhou	36°10'21"N, 103°28'20"W	135.8°/33S
XL3.35	8	K_1	N.Lanzhou	36°10'19"N, 103°28'22"W	126.5°/32S
XL4.55	8	K	E Minhe	36°17'50"N, 102°55'47"E	118.5°/42°S
XL4.56	8	K	E Minhe	36°17'50"N, 102°55'47"E	118.5°/42°S
XL4.57	8	K	E Minhe	36°18'32"N, 102°55'03"E	300.5°/26°N
XL4.58	8	K	E Minhe	36°19'54"N, 102°53'47"E	249.5°/44°N
XL4.59	8	K	E Minhe	36°19'54"N, 102°53'47"E	261.5°/52°N
XL4.60	8	K	E Minhe	36°19'54"N, 102°53'47"E	263.5°/56°N
			Maqin		
M4.33	8	K	W.Maqin	34°30'22"N, 99°57'56"E	272.1°/62°N
M4.34	8	K	W.Maqin	34°30'22"N, 99°57'56"E	*
M4.35	8	K	W.Maqin	34°30'22"N, 99°57'56"E	247.1°/13°N
M4.36	8	K	W.Maqin	34°30'22"N, 99°57'56"E	213.1°/26°N
M4.37	8	K	W.Maqin	34°30'22"N, 99°57'56"E	174.1°/45°S
M4.38	8	K	W.Maqin	34°30'17"N, 99°57'48"E	*
M4.39	8	K	W.Maqin	34°30'17"N, 99°57'48"N	*
M4.40	8	K	W.Maqin	34°30'17"N, 99°57'48"E	*
M4.41	8	K	S.Maqin	34°24'55"N, 100°16'03"E	276.1°/47°N
M4.42	8	K	S.Maqin	34°24'55"N, 100°16'03"E	260.1°/56°N
M4.43	8	K	S.Maqin	34°24'55"N, 100°16'03"E	*
M4.44	8	K	S.Maqin	34°24'20"N, 100°16'12"E	*
M4.45	9	K	S.Maqin	34°24'20"N, 100°16'12"E	264.1°/90°N
			Fenghuoshan-Erdaogou †		
FSH3.06	23	P	Fenghuoshan	34°44'06"N, 92°53'45"E	113.1°/64°S
FSH3.07	16	P	Fenghuoshan	34°42'06"N, 92°53'45"E	71.1°/64°S
FSH3.08	17	P	Fenghuoshan	34°41'06"N, 92°53'45"E	69.1°/39.5°S
FSH3.09	13	P	S.Fengshoushan	34°21'30"N, 92°43'38"E	286.1°/50°N
FSH3.10	16	P	S.Fengshoushan	34°21'30"N, 92°43'38"E	283.1°/50°N
FSH3.11	17	P	S.Fengshoushan	34°21'30"N, 92°43'38"E	276.1°/52°N
FSH3.12	26	P	Erdaogou	34°37'50"N, 92°47'08"E	297.1°/75°S
FSH3.13	10	P	Erdaogou	34°37'50"N, 92°47'08"E	116.1°/69°S
FSH3.14	14	P	Erdaogou	34°37'50"N, 92°47'08"E	113.1°/66°S

N, number of cores sampled; age, K for Cretaceous, K_1 for Early Cretaceous, and P for Paleocene/early Eocene; Bedding is given as strike (dip direction minus 90°) and dip; N, north; E, east; W, west; S, south.

* Variable bedding attitude at the site.

† Sites FSH3.06 and FSH3.10 to FSH3.11 show slight variations in bedding attitude: average strikes and dips are given.

Paleocene(?) or early Eocene [*Smith and Juntao*, 1988; *Leeder et al.*, 1988].

We sampled nine sites of red beds (FSH3.6 to FSH3.14) along the road from Golmud to Tangulashan in the Erdaogou thrust basin complex (34.5°N, 92.7°E), about 25 km south of the Jinsha suture of Late Triassic/Early Jurassic age [*Chang et al.*, 1986; *Matte et al.*, 1996], which separates the Kunlun Block to the north from the Qiangtang block to the south. Our sites therefore belong to the Qiangtang Block. *Lin and Watts* [1988] sampled two localities (P23 and P24) of this formation in the same area, which are discussed in section 3.4. The bedding and geographic coordinates of each site are given in Table 1. The local magnetic deviation computed from comparison of magnetic and solar azimuths is 2.4°±3.9° (n=88 cores), consistent with the declination computed from the IGRF given the large variance.

3. Paleomagnetic Analysis and Results

3.1. Laboratory Techniques and Measurements

All orientations were corrected for local magnetic deviation determined from either IGRF field or magnetic and Sun azimuths measured in the field as discussed above. Standard 2.5-cm-diameter minicores were cut into 2.2-cm-long specimens in the laboratory. All magnetic measurements and demagnetizations were performed in the magnetically shielded room of the Paleomagnetic Laboratory at the Institut de Physique du Globe de Paris (IPGP) / Université de Paris 7 and also in the Magnetism Laboratory of Saint Maur (IPGP). Remanent magnetizations were measured with a Three-component CTF inc. cryogenic magnetometer. Most specimens were thermally demagnetized within laboratory-built furnaces over 15 to 19 temperature steps up to 690°C. Magnetic mineral transformations were monitored during thermal demagnetization by measuring bulk susceptibility after each heating step, using a Molspin susceptibility meter. In order to assist in the identification of magnetic carriers, acquisition of isothermal remanent magnetization (IRM) and Lowrie tests [*Lowrie*, 1990] were performed on a few selected samples. The Lowrie test consists of applying successively decreasing magnetic fields along three perpendicular directions of the sample (1.2, 0.4 and 0.12 T). Artificially magnetized samples are next demagnetized using either alternating field (AF) or thermal thechniques, and the demagnetization of each component provides constraints on the coercivities of different magnetic fractions. Hysteresis loops were determined on a few samples, using a laboratory-made translation inductometer. Attempts to determine Curie points were carried out with a CS-2 Kappabridge high-temperature susceptibility meter. Very low susceptibilities at high temperatures prevented useful interpretation of the results,

which are not shown here. Demagnetization results were plotted as orthogonal vector diagrams [*Zijderveld*, 1967] and also as equal-area projections. Paleomagnetic directions were determined using principal component analysis [*Kirschvink*, 1980] or using the remagnetization planes technique [*Halls*, 1978]. Site mean directions were determined using the statistics of *Fisher* [1953], or in the case of combined directional data and remagnetization circles, the method of *McFadden and McElhinny* [1988]. All interpretations and data processing were carried out using the PaleoMac software package developed at IPGP by J.P. Cogné.

3.2. Xining-Lanzhou Basin (Hekou Group)

Typical examples of IRM acquisition curves are shown in Figure 3a. From the diagrams one can deduce the presence of a low-coercivity magnetic carrier which becomes saturated by about 0.1-0.15 T. At higher field intensities, there is a continuous increase of IRM intensity up to an applied field of 1.2 T (the maximum value which can be attained with our electromagnet) at which saturation is not yet reached. Thermal demagnetization of the three-axis IRM acquired at 1.2, 0.4, and 0.12 T (Figure 3b) shows that a magnetic phase is destroyed at

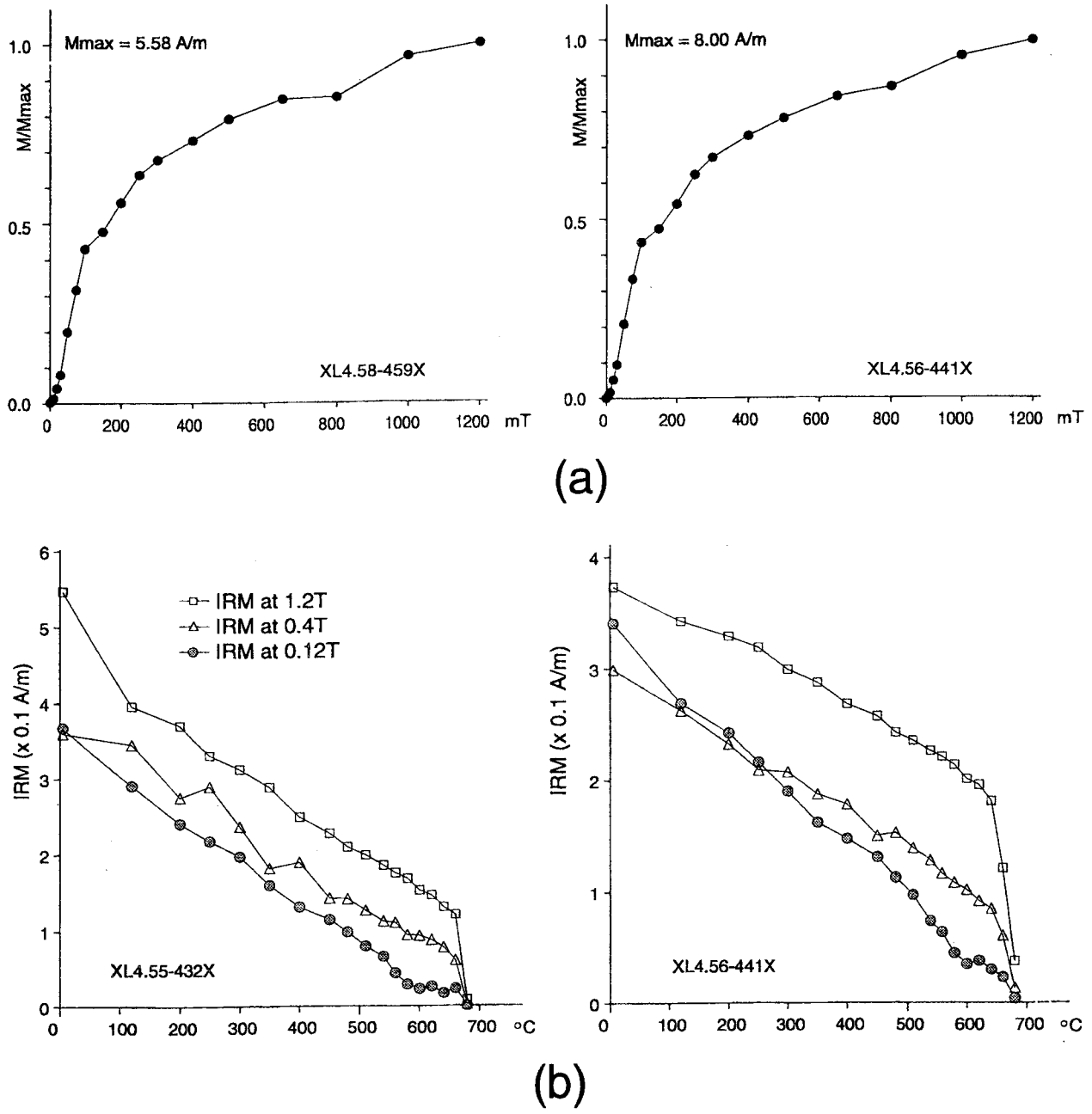


Figure 3. Results of magnetic mineralogy experiments for Xining-Lanzhou basin samples. (a) Isothermal remanent magnetization (IRM) acquisition curves showing the presence of both high- and low-coercivity magnetic minerals. (b) Thermal demagnetization of three axis IRM following *Lowrie* [1990]. (c) Hysteresis loop (top) before and (bottom) after correction of paramagnetic component.

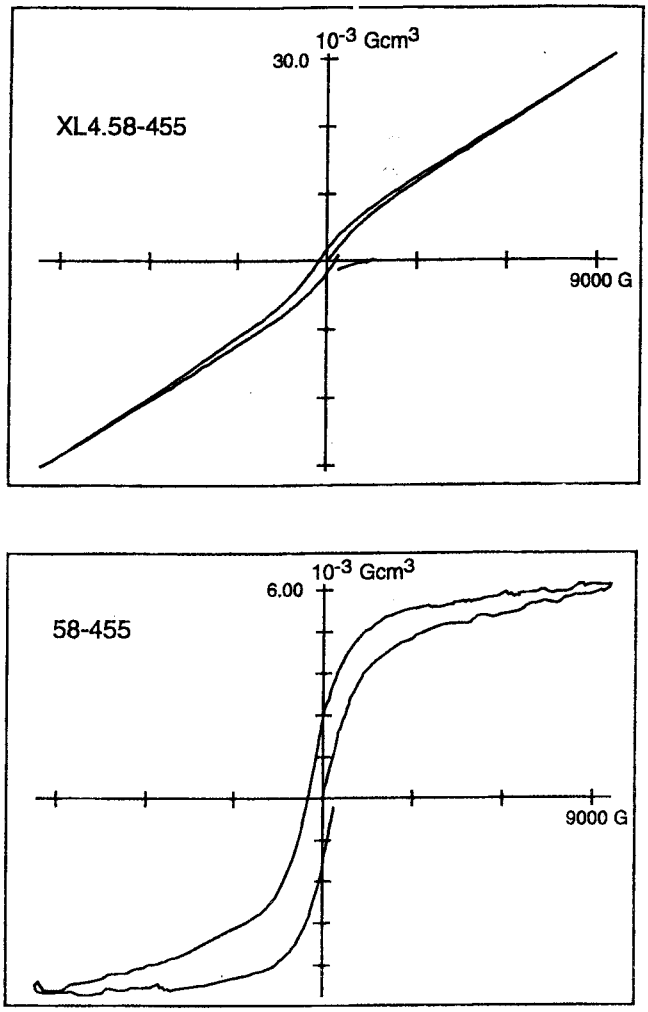


Figure 3. (continued)

580°C. This, together with the saturation of a magnetic carrier around 0.1-0.15 T, indicates the presence of magnetite. The remaining soft fraction disappears at 680°C, together with the medium and hard fractions. Hysteresis loops (Figure 3c) show that samples are not saturated after applying a field of 0.9 T, supporting the presence of a hard magnetic carrier and are generally "wasp-waisted." *Roberts et al.* [1995] interpret this characteristic shape as evidence for a bimodal (or multimodal) population of magnetic carriers, with very different coercivities. This can be due to a single magnetic mineral with a variety of grain sizes or to a combination of magnetic minerals with different coercivities. The H_{cr}/H_c ratio is between 4.2 and 10.4, which suggests a heterogeneous magnetic mineralogy [*Roberts et al.*, 1995]. If we turn to Figure 3a, a kink in the IRM acquisition curves is observed around 0.8 T. Taken together, these experiments suggest that the main magnetic carriers are mainly hematite with some contribution from magnetite. Hematite with large grain sizes would correspond to the medium coercivity and part of the low-coercivity fractions, fine-grained hematite would correspond to the hard fraction, and magnetite would correspond to the remainder of the soft fraction.

Typical thermal demagnetization curves are shown in Figure 4 in tilt-corrected coordinates. These demagnetizations have been the subject of a separate analysis [*Halim et al.*, 1996]. Natural remanent magnetization (NRM) intensity averages 9.9 mA/m. A low unblocking temperature component (LTC) is apparently removed in most samples by 350°C but can persist up to 480°C. IRM acquisition and Lowrie tests described above show that this magnetization is probably carried both by magnetite and larger grain size fractions of hematite. This LTC was identified in 57 specimens (out of 134 treated specimens) from seven sites (31, 33, 35, 55, 58, 59, and 60). The mean direction [*Fisher*, 1953], at the specimen level, for this component is D_g=14.5°, I_g=54.8° (k_g=40.0, α_{95}=3.0°) before tilt correction. The fold test [*McElhinny*, 1964] is negative at the 99% confidence level (with k_g/k_s=6.8, n=57), suggesting that this component is postfolding in age. We therefore assume that it is a recent overprint. One may note, however, that the average LTC declination does not conform to either that of the geocentric axial dipole (GAD) field (D=0°, I=55.7°) or of the present Earth field (PEF) (D=-1.8°, I=52.3°), calculated from coefficients of the 1990 IGRF and their time derivatives. This discrepancy could either be due to a very recent clockwise rotation of about 15° or to a poor separation of the LTC in some samples, where it is not clearly distinct from the following component.

After removal of part of this LTC, magnetization regularly decreases between 480°C and 660°C and then sharply at 680°C to 690°C (Figure 4d). This behavior, which is in good agreement with the observations of the Lowrie experiments (Figure 3b), suggests the presence of both magnetite and hematite as magnetic carriers recording two magnetization components within the 0°C to 580°-600°C and 300°-400°C to 680°C, respectively, unblocking temperature ranges. In most specimens analyzed, the demagnetization paths for both unblocking temperature spectra are linear in orthogonal projections (Figure 4). The high unblocking temperature component (HTC) is unambiguously determined in the highest (>600°C) range. Concerning the magnetization trajectory in the 300°-400°C to 580°-600°C range, it is generally linear and either parallels the HTC at the specimen level (Figure 4a) or makes some angle with it (Figure 4c). In the latter case, an intermediate temperature component (ITC) could be determined in this temperature range. The interpretation of this ITC is discussed in detail by *Halim et al.* [1996], who propose that this component results from overlapping of the unblocking temperature spectra of the LTC and HTC.

The HTC was isolated in most samples, except those from site 57. As observed from the example in Figure 4b, the HTC may trend in a southerly and upward direction, which suggests the presence of a reverse polarity component. This is the only site where we find a reverse polarity, all other sites having normal ones. Unfortunately, because of low intensity, the mean HTC direction for site 57 could not be determined with confidence, despite the use of the remagnetization great circles technique. Site-mean directions of the HTC from the 11 other sites are listed in Table 2 and illustrated on equal-area projections in Figure 5, along with the average HTC directions of localities A and B from Frost et al. (1995). Apart from sites 55 and 56, which appear as outliers and have been excluded from the final mean (our notes in the field book underlined that this outcrop could be structurally displaced and was suspect), it can be seen that the 12 remaining directions from different localities (Xining, site 29; Sunan, locality A; north of Lanzhou,

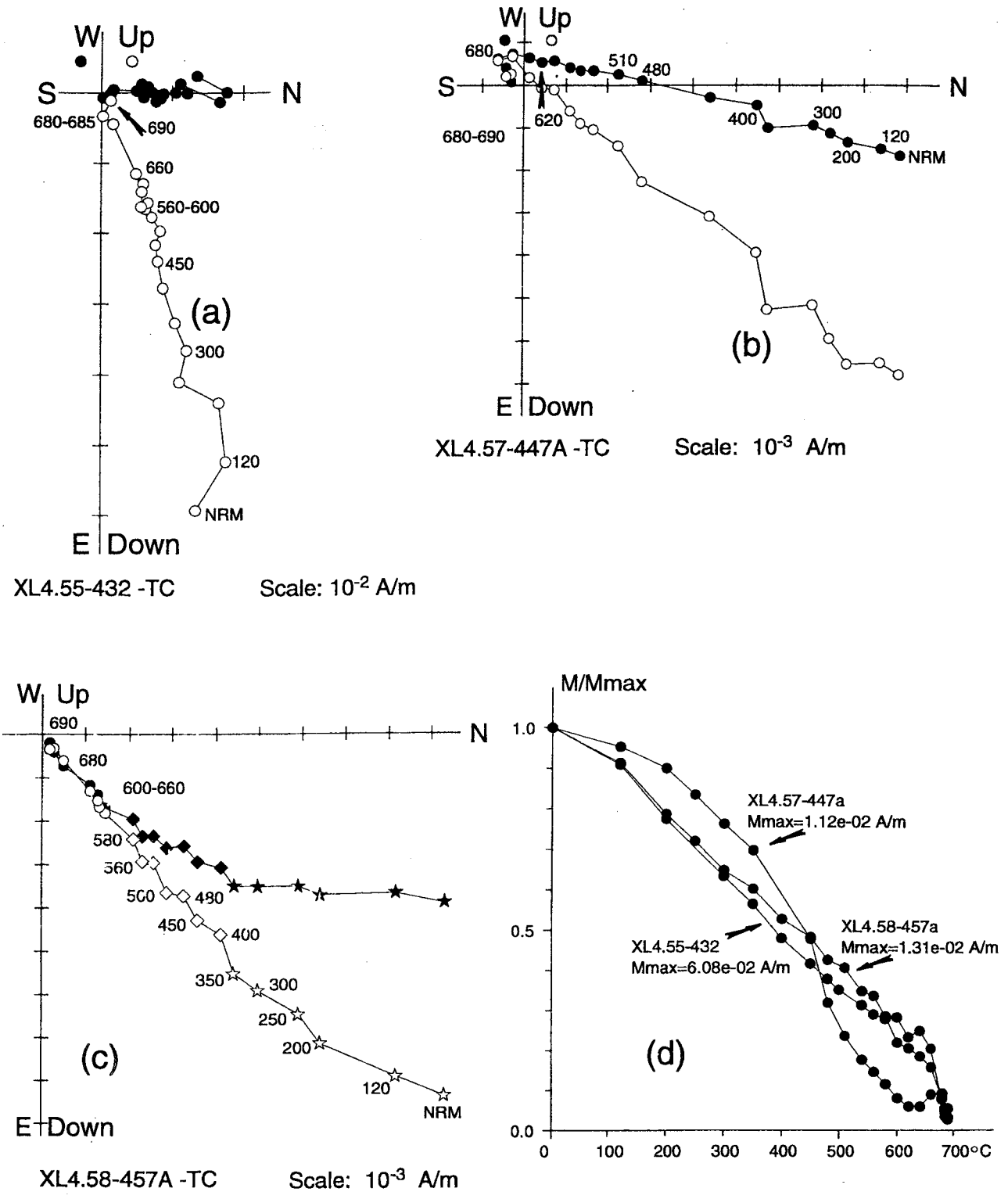


Figure 4. (a), (b), and (c) Typical orthogonal vector plots [*Zijderveld*, 1967], in tilt-corrected coordinates, of three Xining-Lanzhou specimens. Stars (dots), low (high) temperature component; diamonds, apparent intermediate temperature component when it forms an angle with the high temperature component; solid (open) symbols, projection onto the horizontal (vertical) plane. Temperatures are indicated in °C. (d) Normalized magnetization intensity curves of the same specimens as in Figures 4a-4c.

Table 2. Site-Mean Paleomagnetic Directions for the High-Temperature Components (HTC) From the Xining-Lanzhou Basin

Site	n/N	D_g	I_g	D_s	I_s	k	α_{95}	λ_p	ϕ_p	dp/dm (A_{95})	plat
XL3.29	7/8	294.2	81.4	35.8	37.8	94.0	6.3	55.4	207.4	4.4/7.4	21.1
XL3.30	7/8	325.2	78.7	44.7	46.0	69.6	7.3	51.3	193.6	6.0/9.3	27.3
XL3.31	8/8	37.5	38.4	46.8	48.5	98.8	5.6	50.5	189.4	4.8/7.4	29.4
XL3.32	8/8	38.7	46.5	56.1	42.2	30.0	10.3	40.9	190.9	7.8/12.6	24.4
XL3.33	7/8	52.0	-22.3	47.9	51.9	60.6	7.8	50.7	184.3	7.3/10.7	32.5
XL3.34	7/8	36.5	8.2	35.1	41.0	153.1	4.9	57.3	206.4	3.6/5.9	23.5
XL3.35	7/8	28.8	17.6	27.8	49.5	22.8	12.9	66.2	199.6	11.4/17.2	30.3
XL4.55 *	10/10	21.7	15.5	15.9	57.0	45.6	7.2	77.2	182.4	7.6/10.5	37.5
XL4.56 *	6/8	18.2	31.9	-2.0	72.0	307.8	4.1	69.3	99.8	6.4/7.2	56.9
XL4.57	-	-	-	-	-	-	-	-	-	-	-
XL4.58	15/15	89.1	44.9	44.2	42.0	107.7	3.7	50.3	197.9	2.8/4.5	24.2
XL4.59	15/15	98.1	41.2	53.6	34.7	142.7	3.2	40.2	198.0	2.1/3.7	19.1
XL4.60	14/14	115.1	43.8	53.4	44.1	52.6	5.5	43.7	190.2	4.3/6.9	25.8
A †	16	225.8	-44.0	-	-	75.5	4.4	-	-	-	-
A †	16	-	-	221.9	-39.0	55.8	5.1	-50.6	22.6	3.6/6.1	-22.0
B †	5	237.1	-48.6	-	-	5.3	38.7	-	-	-	-
B †	5	-	-	231.8	-44.1	69.0	10.0	-45.0	371.4	7.9/12.5	-25.8
Average											
HTC	12/15	54.2	44.9	-	-	4.9	21.9	-	-	-	-
HTC	12/15	-	-	45.0	43.7	103.1	4.3	50.3	195.5	4.6	25.9

n/N, number of entry in the statistics / number of treated specimens except for the overall average which is based on the site level; D_g,I_g (D_s,I_s) mean directions in geographic, in situ (stratigraphic, tilt-corrected) coordinates; k, α_{95}, parameters of *Fisher* [1953] statistics; λ_p (ϕ_p), latitude (longitude) of the corresponding HTC paleopole. dp/dm (A_{95}), small/great axis (radius) of the ellipse (95% circle) of confidence of the paleopole; plat, Cretaceous paleolatitude of the locality calculated from stratigraphic coordinates.

* Sites 55 and 56 are rejected from the final HTC mean; no HTC could be determined for site 57.

†Sites A and B from *Frost et al.* [1995] included in the final average.

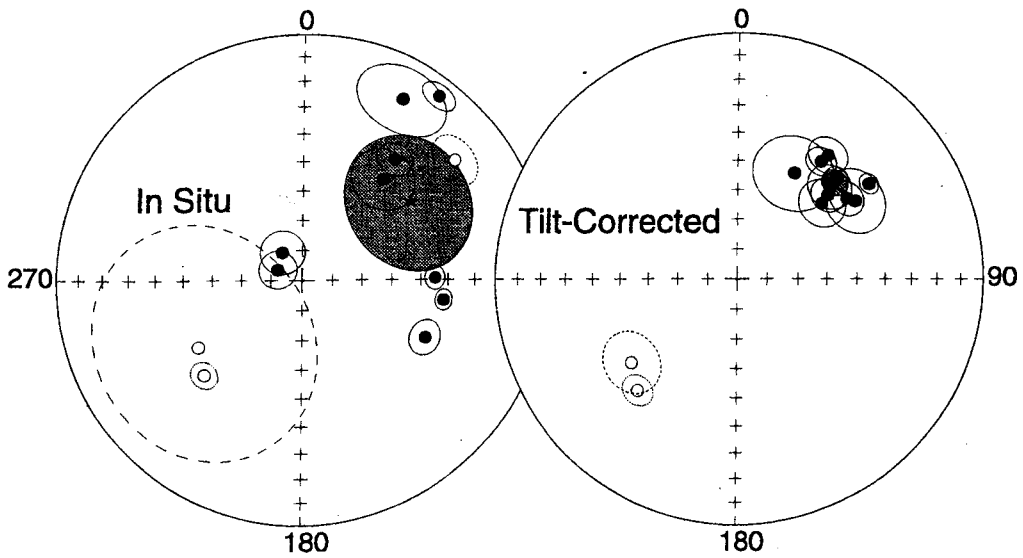


Figure 5. Equal-area projections of site-mean directions with their α_{95} cones of confidence of high-temperature component from Xining-Lanzhou basin before and after bedding correction. Sites A and B from *Frost et al.* [1995]. Solid (open) symbols, positive, downward, (negative, upward) inclinations.

sites 30 to 35; and west of Lanzhou, sites 55 to 60 and locality B) are not significantly different.

Because some sites (A and 29) are situated in widely different locations, we computed an individual paleopole for each site from the corresponding tilt-corrected site-mean HTC (Table 2) and then calculated the paleopole using *Fisher* [1953] statistics. For the same reason, the fold test [*McElhinny*, 1964] was carried out on these paleopoles. This fold test appears to be positive at the 99% confidence level, with k_s/k_g=22.0, n=12. The overall mean paleopole lies at 195.5°E, 50.3°N (A_{95}=4.6°, n=12), giving a paleolatitude of 25.9°±3.5°. The HTC is interpreted as a prefolding primary magnetization, representative of the Lower Cretaceous magnetic field for the Xining-Lanzhou basin.

We next considered whether the computed paleopole is characteristic of the whole Qaidam block. Site A, which was sampled on the northern side of the Qilian Shan, gives the same results as other sites (site B of *Frost et al.* [1995], and all of our sites) which have been sampled along the southern border. In the following, the calculation of relative and absolute movements deduced from these results is attributed to the Xining-Lanzhou basin; extending the conclusions to the Qaidam block is suggested below but would require further paleomagnetic studies more to the west, in the interior of the block.

3.3. Maqin Section (Wanxiou Group)

A representative example of IRM acquisition is presented in Figure 6a. Samples seem to acquire a weak magnetization in the low-field range and do not saturate at high applied fields (1.2 T). These experiments could suggest the presence of a low-coercivity mineral, as shown by the kink in the IRM acquisition curve between 0.15 and 0.2 T, and of a high-coercivity magnetization carrier that appears to be hematite (total unblocking of the HTC occurs at 680°C; Figure 4c). NRM intensity is 14.0 mA/m.

Two typical demagnetization plots of samples from the Maqin region are presented in Figures 6c and 6d, which show that a weak LTC is removed at 520°-580°C in the west Maqin samples and persists up to 630°C in the south Maqin samples. The LTC from the south and west Maqin areas has been computed at the sample level: its average direction is D_g=3.9°, I_g=40.0° (k_g=26.7, α_{95}=3.8°, n=55) in geographic coordinates; clustering of the directions is better before bedding correction than after (with k_g/k_s =7.6) demonstrating a postfolding acquisition. Thus this component is likely a recent remagnetization. However, the inclination of the LTC is lower than both the present Earth field (PEF) direction (D=-0.9°, I=51.9°; IGRF, coefficients of 1990), and the GAD field (D=0°, I=54°) at the sampling locality. This could be due to a poor separation of magnetization components, arising from the weak intensity of LTC, or the low number of demagnetization steps used to resolve it.

At higher temperatures, the demagnetization path trends toward the origin. Total unblocking of the HTC occurs at 680°C, which suggests that the magnetic carrier is hematite. In the example of Figure 6d, the curve does not converge toward the origin but tends to reach a southerly upward, reverse, direction. In this case, points align on a great circle because of overlap between LTC and HTC (Figure 6b). The HTC exhibits these two different behaviors depending on the location of the sites. In the west Maqin area (sites M4.33-40) the HTC could be determined using principal component analysis, whereas in the south Maqin sites (M4.41-45) the poor separation of components allowed only the application of the remagnetization great circles technique. Demagnetization paths always converge toward the south with negative inclinations in stratigraphic coordinates (Figures 6b and 6d), and we thus assume that the southern hemisphere intersection of the great circles defines the true HTC direction.

Site-mean directions of the HTC are shown in Figure 7 and listed in Table 3. The data are much more scattered before unfolding than after. Because the polarities from the west Maqin samples are entirely normal and those from south Maqin all

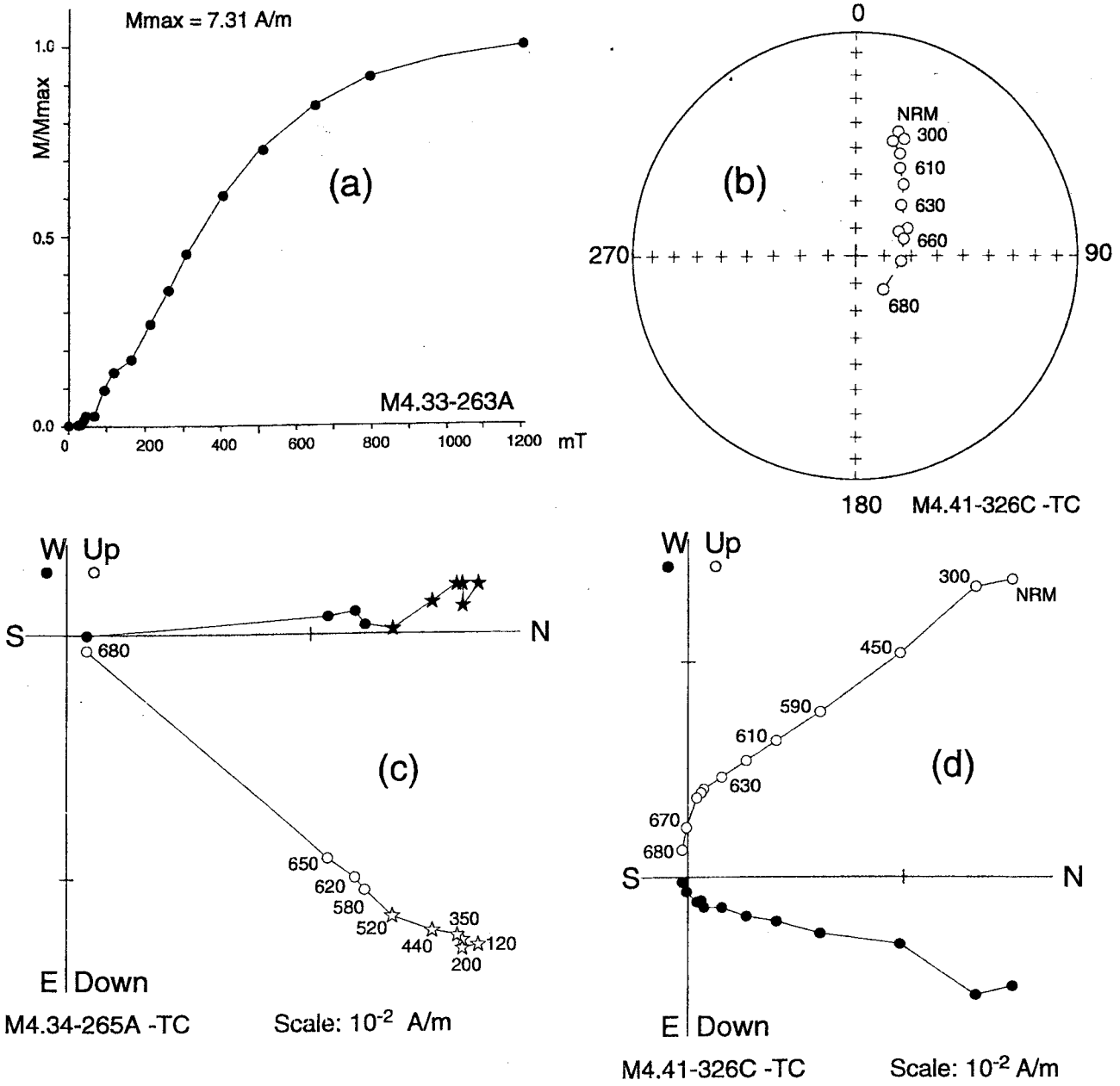


Figure 6. Thermal demagnetization of samples from Maqin area. (a) Normalized IRM acquisition curve. (b) Example of evolution of the magnetization direction along a great circle during thermal demagnetization (temperatures in °C). (c) and (d) Typical orthogonal vector plots of specimens from west and south Maqin collections. Same conventions as in Figure 4.

reverse, we analyze these two populations separately. For the west Maqin sites, the formation mean direction after tilt correction is D_s=352.1°, I_s=40.2° (k_s=233.0, α_{95}=3.6°), and the fold test is positive at the 99% level of confidence with k_s/k_g=38.2, much higher than the critical value (3.7) for n=8. The magnetization is therefore likely prefolding in age. Turning to the south Maqin sites, the average HTC direction is D_s=194.9°, I_s=-45.3° (k_s=7.5, α_{95}=29.9°). Although the data also cluster better upon unfolding (Figure 7), the fold test is inconclusive at the 95% level (k_s/k_g=3.41, n=5, critical value of 3.44). This situation arises because of the anomalous inclination (in stratigraphic coordinates) of site 41 (Figure 7). If we remove this site from computations, the fold test becomes positive at the 99% level of confidence (k_s/k_g=11.50, n=4, critical value of 8.47). We therefore assume that south Maqin directions are prefolding.

We face three problems with the south Maqin sites: (1) the fold test is not robust, (2) directions have been determined only from great circle intersections, which could be, in part, responsible for the large remaining dispersion of the site-mean directions after tilt correction, and (3) all the data from west Maqin are normal, while those from south Maqin are reverse. The possibility that south Maqin red beds are not of the same age as in west Maqin should be considered. Yet, if we compute the mean direction from all sites from both south and west Maqin (Table 3), they pass the fold test at 99% confidence limits, and the mean direction after tilt correction is not significantly different from the west Maqin average (but the

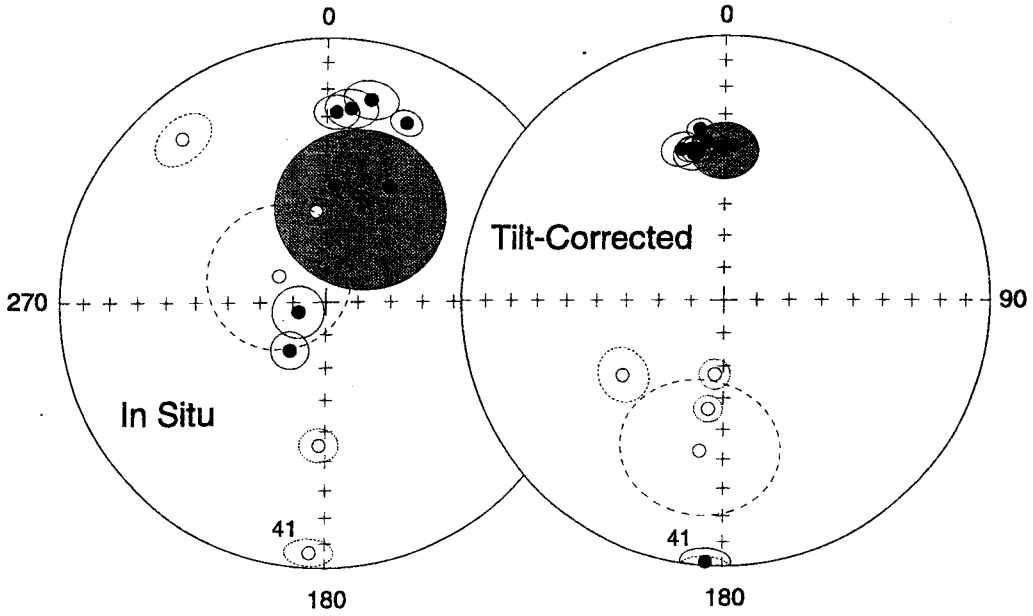


Figure 7. Equal-area projection of site-mean directions with their α_{95} ellipses of confidence of high-temperature component from Maqin region before and after bedding correction. Same conventions as in Figure 5.

cone of confidence is much larger). For these reasons, we retain both the west and south Maqin site means to compute the formation mean direction for the Lower Cretaceous Wanxiou group, which is D_s=359.8°, I_s=42.5° (k_s=17.2, α_{95}=10.3°). The corresponding pole lies at 281.2°E, 80.1°N (dp=7.8°, dm=12.7°). This pole (Table 5) implies a Cretaceous paleolatitude of 24.6°±6.2° for the region.

3.4. Erdaogou Basin (Fenghuoshan Group)

Typical examples of thermal demagnetization of specimens from the Fenghuoshan region are given in Figure 8. The curves of magnetic moment evolution versus temperature (Figure 8a) generally show little variation between NRM and 300°C to 400°C; only a few specimens show a decrease in intensity in this temperature interval. An inflexion point is observed around 580°C, followed by a sharp fall off around 680°-690°C. These observations suggest the presence of at least two different magnetic carriers within the 300°-600°C and 600°-690°C unblocking temperature spectra. The magnetic carrier corresponding to the second temperature interval is hematite, but the one corresponding to the first interval is more difficult to define and could be either magnetite or hematite in large grain sizes. Bulk susceptibility (Figure 8b) remains constant during heating until 600°C, then sharply increases, suggesting alteration at high temperatures (>600°C).

Two vector diagrams are shown in Figures 8c and 8d, which illustrate the general behaviour of most samples. Both Tertiary and Cretaceous samples are weakly magnetized (NRM around 2.5 mA/m). No LTC could be isolated below 350°C. Above 400°C, two kinds of demagnetization paths were obtained. In the first (Figure 8c), the path converges toward the origin, allowing the resolution of a single HTC component in the 580°-680°C range. This is the case for most samples. In the second, particularly for samples from site 13, the envelope of the vector end-points misses the origin (Figures 8d and 8e), and the vector end points align on a great circle (Figure 8f) between 250°-350°C and 640-680°C. For the latter case, we attempted to resolve the HTC using the remagnetization great circle technique. In some samples, demagnetizations were too noisy to isolate magnetization components with any confidence. All samples from site 09 displayed this behavior, and this site had to be discarded in the following.

The HTC site-mean directions (Table 4 and Figure 9) were determined using *Fisher* [1953] statistics in the case of directional data (sites 7 and 8) or *McFadden and McElhinny* [1988] statistics in the case of mixed directional data and remagnetization great circles (all other sites). The data are much better grouped after tilt correction. However, the site 6 mean direction lies away from the others after tilt correction. This may be due to either erroneous measurement of the bedding attitude in the field or a locally rotated site. This site has therefore been excluded from the final formation mean. The k_s/k_g ratio is 34.6 which is significantly higher than the critical value (4.16) for n=7. Thus the fold test is positive at the 99% level of confidence [*McElhinny*, 1964]. Moreover, the HTC passes the reversal test [*McFadden and Lowes*, 1981]. For these reasons the prefolding magnetization is likely the primary magnetization for this formation.

As mentioned above, *Lin and Watts* [1988] performed a paleomagnetic study of the same formation at two localities. The structural attitudes at the sites did not allow the application of the fold test. Locality P23 samples displayed a ChRM clustering around the present-day dipole field direction in in situ coordinates, with normal polarities only. Because of the absence of the fold test, we follow *Lin and Watts* , who suspect that the ChRM could be a recent remagnetization, and do not consider these directions. In contrast, sites from locality P24 show a reverse direction in the 350°-600°C unblocking temperature range, with a tilt-corrected direction of D_s=215°, I_s=-34° (k_s=79, α_{95}=4.9°), close to the one we obtained (D_s=9.5°, I_s=34.3°, k_s=47.0, α_{95}=8.9°). We therefore computed a single average of our 7 sites and the 12 sites of locality P24 of

Table 3. Site-Mean Paleomagnetic Directions for the High-temperature Components (HTC) from Maqin Region

Site	n/N	D_g	I_g	D_s	I_s	k	α_{95}	λ_p	ϕ_p	dp/dm	plat
M4.33	6/8	216.8	71.7	348.1	42.4	141.4	5.7				
M4.34	6/7	250.6	81.1	-	-	72.9	7.9				
M4.34	6/7	-	-	351.8	34.6	347.8	3.6				
M4.35	7/7	4.1	53.2	358.3	41.3	387.2	3.1				
M4.36	8/8	28.5	48.8	3.2	40.8	716.4	2.1				
M4.37	6/8	23.9	26.7	353.9	39.2	231.4	4.4				
M4.38	6/8	2.9	28.4	-	-	122.6	6.1				
M4.38	6/8	-	-	344.1	40.1	138.4	5.7				
M4.39	8/8	12.3	22.4	-	-	63.3	7.0				
M4.39	8/8	-	-	349.9	41.3	147.2	4.6				
M4.40	8/8	7.2	26.7	-	-	63.5	7.0				
M4.40	8/8	-	-	347.4	40.8	164.3	4.3				
M4.41	9/9	182.6	-45.3	183.7	1.7	109.5	5.6				
M4.42	14/14	299.6	-73.6	-	-	4.1	22.3				
M4.42	14/14	-	-	188.9	-42.3	4.0	22.7				
M4.43	13/13	353.5	-61.6	187.9	-56.2	111.3	4.2				
M4.44	6/8	183.6	-6.3	-	-	106.4	5.6				
M4.44	6/8	-	-	186.6	-67.2	154.2	4.6				
M4.45	10/11	319.2	-18.6	233.6	-51.0	33.4	8.1				
Averages											
All	13/13	20.3	58.5	-	-	3.6	25.7	-	-	-	-
All	13/13	-	-	359.8	42.5	17.2	10.3	80.1	281.2	7.8/12.7	24.6
West Maqin	8/13	9.6	49.9	-	-	6.1	24.3	-	-	-	-
West Maqin	8/13	-	-	352.1	40.2	233.0	3.6	76.5	313.0	2.6/4.3	22.9
South Maqin	5/13	250.2	-70.1	-	-	2.2	69.2	-	-	-	-
South Maqin	5/13	-	-	194.9	-45.3	7.5	29.9	-75.1	37.1	24.1/37.9	26.8

n/N, number of entry in the statistics / number of treated specimens except for the overall average which is based on the site level; D_g,I_g (D_s,I_s) mean directions in geographic, in situ (stratigraphic, tilt-corrected) coordinates; k, α_{95}, parameters of *Fisher* [1953] statistics; λ_p (ϕ_p), latitude (longitude) of the corresponding HTC paleopole; dp (dm), small (great) axis of the ellipse of confidence of the paleopole; plat, Cretaceous paleolatitude of the locality calculated from stratigraphic coordinates.

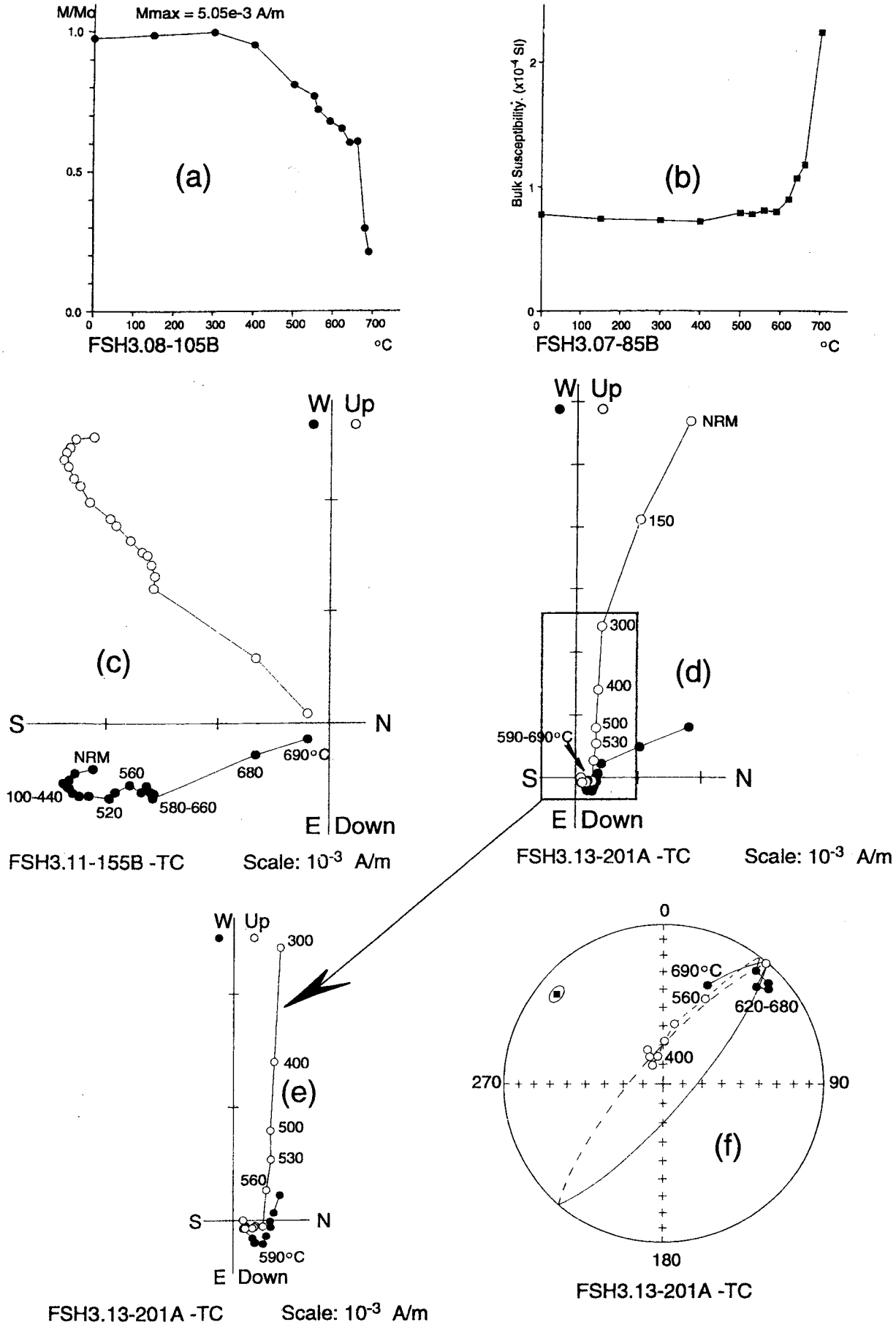


Figure 8. Thermal demagnetization of samples from Fenghuoshan-Erdaogou basin. (a) Normalized NRM decay curve. (b) Bulk suceptibility (in volumetric SI units) evolution with increasing temperature steps. (c) and (d) orthogonal projections of two typical specimens; same conventions as in Figure 4. (e) Details of the magnetization evolution in the high-temperatures steps for the specimen shown in Figure 8d. (f) Evolution along a great circle of the magnetization direction of the specimen shown in Figure 8d.

Lin and Watts [1988]. The bedding tilt-corrected formation mean direction is D_s=25.5°, I_s=34.6° (k_s=32.7, α_{95}=6.0°, n=19). The fold test [*McElhinny*, 1964] remains positive at the 99% level, with k_s/k_g=14.9. We thus consider this direction as the Paleocene-early Eocene primary paleomagnetic direction of the Fenghuoshan-Erdaogou group, representative of the east Qiangtang block. The corresponding paleopole, computed from the formation mean HTC, lies at 210.5°E, 62.6°N (dp=3.9°, dm=6.8°). The Fenghuoshan Early Tertiary paleolatitude is 19.0°±3.1° (Table 5).

4. Discussion

4.1. Paleopole Positions

Our new results should next be compared to those from adjacent Eurasian blocks (i.e., North China, South China, Lhasa, West Qiangtang, Markam, Indochina, Tarim, Turfan, Junggar, Fergana, and Siberia; Figure 1). The corresponding paleopoles are listed in Table 5 and shown in Figure 10. Because all these regions are thought to have been accreted by the Cretaceous [*Enkin et al.*, 1992; *Metcalfe*, 1993; *Matte et al.*, 1996], and most of these poles are Cretaceous in age, discrepancies between them and the reference apparent polar wander path (APWP) for Eurasia [*Besse and Courtillot*, 1991] would imply intracontinental deformation (relative latitudinal displacements and rotations) since the Cretaceous, and the most reasonable common cause for this deformation is the collision of India.

These relative north-south movements are given in Table 5 in terms of northward convergence with respect to Eurasia, computed as the difference between the observed paleolatitude of a given sampling area and the expected paleolatitude of this area calculated from the Eurasian reference APWP of *Besse and Courtillot* [1991]. The Early Tertiary pole of Fenghuoshan is compared to the 60 Ma pole of the reference APWP. Cretaceous paleopoles (Xining-Lanzhou and Maqin poles) are compared to the 130-65 Ma reference pole of *Besse and Courtillot* [1991]. The rather long period over which this average is computed may seem undesirable. However, the stability of Eurasian paleopoles from the Late Jurassic to the early Tertiary has long been noted [*Besse and Courtillot*, 1991] and appears to have been confirmed by several studies from Asian blocks: the South China Block [*Enkin et al.*, 1991a, b], Junggar [*Chen et al.*, 1991], Tarim [*Cogné et al.*, 1995], and Tibet [*Chen et al.*, 1993a]. This stability in a way compensates the adverse effect of dating uncertainties that do not allow finer subdivision of most of our Cretaceous samples.

We have computed a new Cretaceous NCB pole, because the previous determination by *Chen et al.* [1993b] was partly based on results obtained in Mongolia [*Pruner*, 1988; *Zhao et al.*, 1990]; yet recent results obtained in the northern part of the Mongol block suggest a nonrigid behavior of Mongolia with respect to NCB (our own unpublished data). The NCB pole is based on Cretaceous data obtained by *Ma et al.* [1993], *Wu et al.* [1993], *Zheng et al.* [1991], and *Gilder and Courtillot* [1997]. Because of the distance between the different sites, we computed an individual paleopole for each published site and then calculated a Fisherian average. This is located at 76.5°N,

Table 4. Site-Mean Paleomagnetic Directions for the High-Temperature Components (HTC) From the Fenghuoshan-Erdaogou Basin

Site	n/N	D_g	I_g	D_s	I_s	k	α_{95}	λ_p	ϕ_p	dp/dm	plat
FSH3.06 *	5/8	337.0	-23.3	-	-	83.7	8.7				
FSH3.06 *	5/8	-	-	337.9	21.2	88.4	8.4				
FSH3.07	7/8	17.5	-27.1	17.2	26.2	38.5	9.9				
FSH3.08	7/8	180.9	14.3	181.9	-23.2	12.7	17.6				
FSH3.10	4/8	196.5	0.5	-	-	61.2	13.7				
FSH3.10	4/8	-	-	194.6	-49.4	60.3	13.8				
FSH3.11	4/8	258.2	80.1	-	-	89.4	10.4				
FSH3.11	4/8	-	-	358.0	41.3	104.9	9.6				
FSH3.12	4/8	230.2	27.3	-	-	169.1	8.2				
FSH3.12	4/8	-	-	7.4	42.4	154.9	8.6				
FSH3.13	5/5	221.6	39.4	-	-	13.8	24.0				
FSH3.13	5/5	-	-	17.9	27.7	12.8	25.1				
FSH3.14	7/8	225.0	34.6	8.9	28.8	17.7	16.1				
Average											
HTC	7/8	306.3	48.5	-	-	1.4	-	-	-	-	-
HTC	7/8	-	-	9.5	34.3	47.0	8.9	62.6	210.5	3.9/6.8	19.0

n/N, number of entry in the statistics / number of treated specimens except for the overall average which is based on the site level; D_g,I_g (D_s,I_s) mean directions in geographic, in situ (stratigraphic, tilt-corrected) coordinates; k, α_{95}, parameters of *Fisher* [1953] statistics; λ_p (ϕ_p), latitude (longitude) of the corresponding HTC paleopole; dp (dm), small (great) axis of the ellipse of confidence of the paleopole; plat, early Tertiay paleolatitude of the locality calculated from stratigraphic coordinates.
* Site 06 is rejected from the final HTC mean.

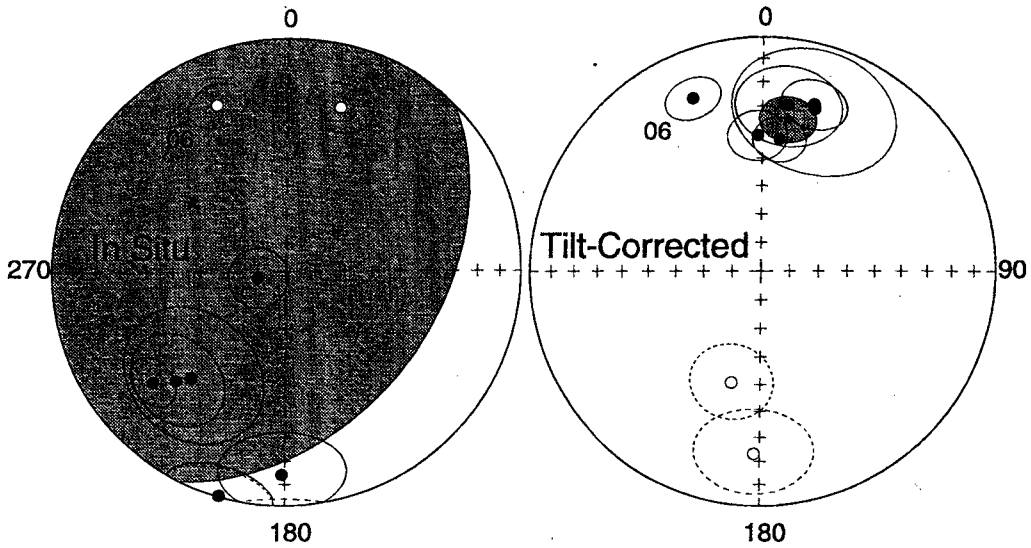


Figure 9. Equal-area projection of site-mean directions with their α_{95} ellipses of confidence of the HTC from Fenghuoshan-Erdaogou basin before and after bedding correction. Same conventions as in Figure 5.

198.0°E (A_{95}=5.0°). The new NCB pole (Figure 10) is virtually identical to both the Eurasian Cretaceous paleopole [*Enkin et al.*, 1992] and the Cretaceous part of the reference APWP of *Besse and Courtillot* [1991], confirming that NCB did not experience any rotation or latitudinal displacement relative to Eurasia since the Cretaceous.

Several important observations can be drawn from the data listed in Table 5 and illustrated in Figure 10. The first is that the Xining-Lanzhou and Maqin paleopoles lie on the same small circle centered on Maqin (Figure 10). This means that no significant relative north-south displacement has taken place between these two places since the Cretaceous. However, the Xining-Lanzhou basin shows a large clockwise rotation (CW, 31.5°±5.5°), and the Maqin area shows a large counterclockwise rotation (CCW, 15.8°±8.9°) with respect to Eurasia (Table 5). Our observations are consistent with the idea that the Kunlun block and Xining-Lanzhou basin were already accreted by the Cretaceous, but this assemblage was subsequently deformed under the effect of the Indian collision since the Tertiary. These modifications are mainly reflected in large relative rotations, which could be associated with smaller block rotation or shear on large fault systems. Pre-Cretaceous collision has been evidenced by the hornblende and whole rock $^{40}Ar/^{39}Ar$ dates and zircon U/Pb dates calculated by *Arnaud* [1992] and *Matte et al.* [1996], indicating that the Kunlun block collided with the Tarim block in Silurian times.

On the other hand, the position of the Early Tertiary paleopole of Fenghuoshan implies a significant northward convergence of the Qiangtang block (15.4°±4.4°) with respect to Eurasia. This value, obtained in the east of Qiangtang, is identical to that found for the Cretaceous (15.1°±4.4°) by *Chen et al.* [1993a] at Domar, in the west of Qiangtang. The fact that these values are identical for Cretaceous (Domar) and Early Tertiary (Fenghuoshan) shows that convergence of Qiangtang with Eurasia took place after the early Tertiary, most likely again under the influence of the Indian collision. Here too, rotations are observed between the western (at Domar, 6.6±5.7° CCW with respect to Eurasia) and the eastern (at Fenghuoshan, 11.5±6.2 CW) parts. This confirms the observation made by *Chen et al.* [1993b] in the Lhasa block, that the Tibetan blocks (Lhasa and Qiangtang) have suffered significant internal deformation under the effect of the collision.

4.2. A Cretaceous Reconstruction

The amounts of convergence of the different blocks with respect to Eurasia (Table 5) match the scheme drawn by *Chen et al.* [1993b]: the total amount of convergence of India (2700 km) since the onset of its collision with Asia is distributed within the Asian plate, with about 1500 km of convergence north of Tibet [*Besse et al.*, 1984], ~1000 km north of Tarim, and ~600 km north of Junggar. We propose a new reconstruction of Cretaceous Asia (Figure 11), including the new results from Qaidam and Kunlun blocks and incorporating recent results from the Shan-Thai-Malay block [*Funahara et al.*, 1993; *Huang and Opdyke*, 1993], and Myanmar area (Kalaw redbeds [*Richter et al.*, 1993]). In this reconstruction, present-day contours of blocks are drawn, with relative positions of the different sampling areas placed at their approximate Cretaceous paleolatitudes (with uncertainties indicated). The general outline of this Cretaceous reconstruction is similar to the one presented by *Chen et al.* [1993b]. However, some uncertainties concerning the paleoposition of Qaidam, Kunlun, and Indochina are partly removed, and some further discussions are warranted.

It is readily seen that the paleolatitude of the Xining-Lanzhou basin (if taken to be part of the Qaidam block) is compatible (within uncertainties), with that from the Tarim and Kunlun sites: the predicted paleolatitude at the Xining-Lanzhou area from attachment to a rigid Tarim block is 23.3±2.5° versus the observed 25.9±3.5°. The Altyn-Tagh fault, located between Tarim and Qaidam, has undergone some 700 km of left-lateral motion over the last 15 Myr [e.g., *Meyer et al.*, 1998].

Table 5. Selected Paleopoles Used for Reconstructing the Cretaceous Asian Mosaic

Pole	Block	Area	Age	Site		Paleopole			Observed Paleolat.	Expected Paleolat.	Convergence /Eurasia	Rotation /Eurasia	References
				Lat	Lon	Lat	Lon	dp/dm (A95)					
	Eurasia (APWP)												
1	130-65Ma	-		-	-	76.3	202.6	2.2					*Enkin et al.* [1992]
2	60Ma	-		-	-	78.5	178.7	3.9					*Besse and Courtillot* [1991]
	Mongolia												
3			K	46.8	107.3	86.2	337.5	6.9/8.8	44.3±5.5	43.9±1.8	-0.4±5.8	23.2±10.2	(12) *Pruner* [1988]
4			K	46.8	107.3	80.9	12.7	7.7/9.7	45.3±6.2	43.9±1.8	-1.4±6.4	32.1±11.4	(13) *Pruner* [1988]
5			K	43.8	107.9	72.5	204.5	6.4/8.6	39.4±5.1	41.2±1.8	1.8±5.4	-4.5±8.8	(14) *Pruner* [1988]
6			K_1	48.9	117.0	82.6	247.3	6.7	43.8±5.4	47.0±1.5	3.2±5.6	15.3±9.7	(J1) *Zhao et al.* [1990]
7			K_1	48.9	117.0	77.1	358.2	5.3	41.6±4.2	47.0±1.5	5.4±4.4	37.7±7.6	(J2) *Zhao et al.* [1990]
8			K_1	42.0	119.2	84.8	258.4	8.3	38.0±6.6	41.2±1.5	3.2±6.8	16.1±10.9	(A7-A10) *Zhao et al.* [1990]
9			K_1	53.8	124.5	60.1	48.8	4.2/5.1	50.5±3.4	53.4±1.5	2.9±3.7	75.1±7.3	*Halim et al.* [1998]
10			K_1	53.8	124.5	56.0	55.0	6.5/7.7	50.9±5.2	53.4±1.5	2.5±5.4	83.3±10.9	*Halim et al.* [1998]
	China												
11	North China	*	K	40.0	110.0	76.5	198.0	5.0	39.1±4.0	38.0±1.8	-1.1±4.4	0.0±7.0	this study from *Ma et al.* [1993], *Wu et al.* [1993], *Zheng et al.* [1991] and *Gilder and Courtillot* [1997]
12	South China	*	K	28.0	110.0	76.5	203.8	4.4	26.3±3.5	26.5±1.8	0.2±3.9	-0.2±5.4	*Lin* [1984], *Kent et al.* [1987], *Zhu et al.* [1988], *Hu et al.* [1990], *Chan* [1991], *Huang and Opdyke* [1992], *Chen et al.* [1993b]
	Central Asia												
13	Khazakhstan	Fergana	K	40.5	72.5	74.3	276.0	5.4/8.6	25.9±4.3	31.0±1.8	5.1±4.6	19.1±6.5	*Bazhenov* [1993]
14	Junggar	Manas	K	44.2	86.0	73.4	223.6	4.3	31.1±3.4	37.0±1.8	5.9±4.6	2.4±5.7	*Chen et al.* [1991]
15	Tarim	Yingjisha	K	38.5	76.4	71.2	226.7	4.7/8.0	21.7±3.7	29.7±1.8	8.0±4.1	2.1±9.5	*Chen et al.* [1992]
16		Wuqia	K	39.5	75.0	70.1	225.8	4.9/8.2	21.6±3.9	30.4±1.8	8.8±5.1	2.3±5.9	*Chen et al.* [1992]
17		Kuche	K	41.6	83.5	66.2	223.3	6.1/10.3	22.1±4.8	34.0±1.8	11.9±5.7	1.8±8.9	*Li et al.* [1988]
18		Turfan	K	43.0	90.0	65.1	227.8	5.1/8.4	23.0±4.0	36.6±1.8	13.6±5.0	2.1±7.6	*Cogné et al.* [1995]
		Average *		41.0	80.0	68.4	224.5	3.1	22.6±2.5	32.8±1.8	10.2±3.1	0.3±4.3	*Cogné et al.* [1995]
	Tibet												
19	Qaidam	Xining-Lanzhou	K_1	36.2	103.5	50.3	195.5	4.6	25.9±3.5	32.9±1.8	7.0±4.6	-31.5±5.5	*Frost et al.* [1995] and this study
20	Kunlun	Maqin	K_1	34.5	100.1	80.1	281.2	7.8/12.7	24.6±6.2	30.5±1.8	5.9±6.4	15.8±8.9	*this study*
21	Qiangtang	Domar	K	33.75	80.4	66.2	245.0	5.1	10.7±4.0	25.8±1.8	15.1±5.0	6.6±5.7	*Chen et al.* [1993a]
22		Fenghuoshan	E_1	34.5	92.8	62.6	210.5	3.9/6.8	19.0±3.1	34.5±3.1	15.4±4.4	-11.5±6.2	*Lin and Watts* [1988] and this study
23	Lhasa	S Lhasa	K	30.0	91.0	69.1	283.5	2.8/5.4	9.6±2.2	24.2±1.8	14.6±2.9	18.5±6.2	*Achache et al.* [1984], *Lin and Watts* [1988], *Chen et al.* [1993b]
24		N Lhasa	K	31.5	91.3	63.4	326.2	5.9	14.3±4.7	25.7±1.8	11.4±5.0	36.3±6.8	*Pozzi et al.* [1982], *Achache et al.* [1984], *Lin and Watts* [1988], *Chen et al.* [1993b]
25	East Tibet	Markam	K	29.7	98.6	48.6	173.5	6.0	31.3±4.8	25.6±1.8	-5.7±5.7	-33.6±7.4	*Otofuji et al.* [1989] *and Huang and Opdyke* [1992]
	Indochina												
26		Khorat	J+K	16.5	103.0	63.8	175.6	1.7	22.4±1.3	13.8±1.8	-8.6±3.3	-15.2±2.9	*Yang and Besse* [1993]
27		Kalaw	J+K	20.8	96.4	55.9	195.0	6.1/11.4	12.4±4.9	16.5±1.8	4.1±5.2	20.9±6.6	*Richter et al.* [1993]
28		Simao (Xiaguan)	K_2	25.6	100.2	83.1	161.2	7.3/11.2	28.8±5.8	22.0±1.8	-6.8±6.0	-7.5±8.6	*Huang and Opdyke* [1993] *and Funahara et al.* [1993], *Yang et al.* [1995]
29		Simao (Jinggu)	K_{1-2}	23.4	100.9	18.8	170.8	7.0/11.3	25.2±5.6	20.0±1.8	-5.2±5.9	65.4±8.0	
30		Simao (Mengla)	K_{1-2}	21.6	101.4	33.8	179.7	5.3/9.0	21.2±4.2	18.4±1.8	-2.8±4.6	46.6±6.1	

Pole, pole number; J, Jurassic, K, Cretaceous, K_1, Early Cretaceous, and K_2, Late Cretaceous; site (lat, lon), latitude and longitude of sites; paleopole (lat, lon), latitude and longitude of paleopoles.
* Site coordinates for NCB, SCB and Tarim average are arbitrary reference coordinates within these blocks. Calculations according to *Coe et al.* [1985] for errors in paleolatitude.

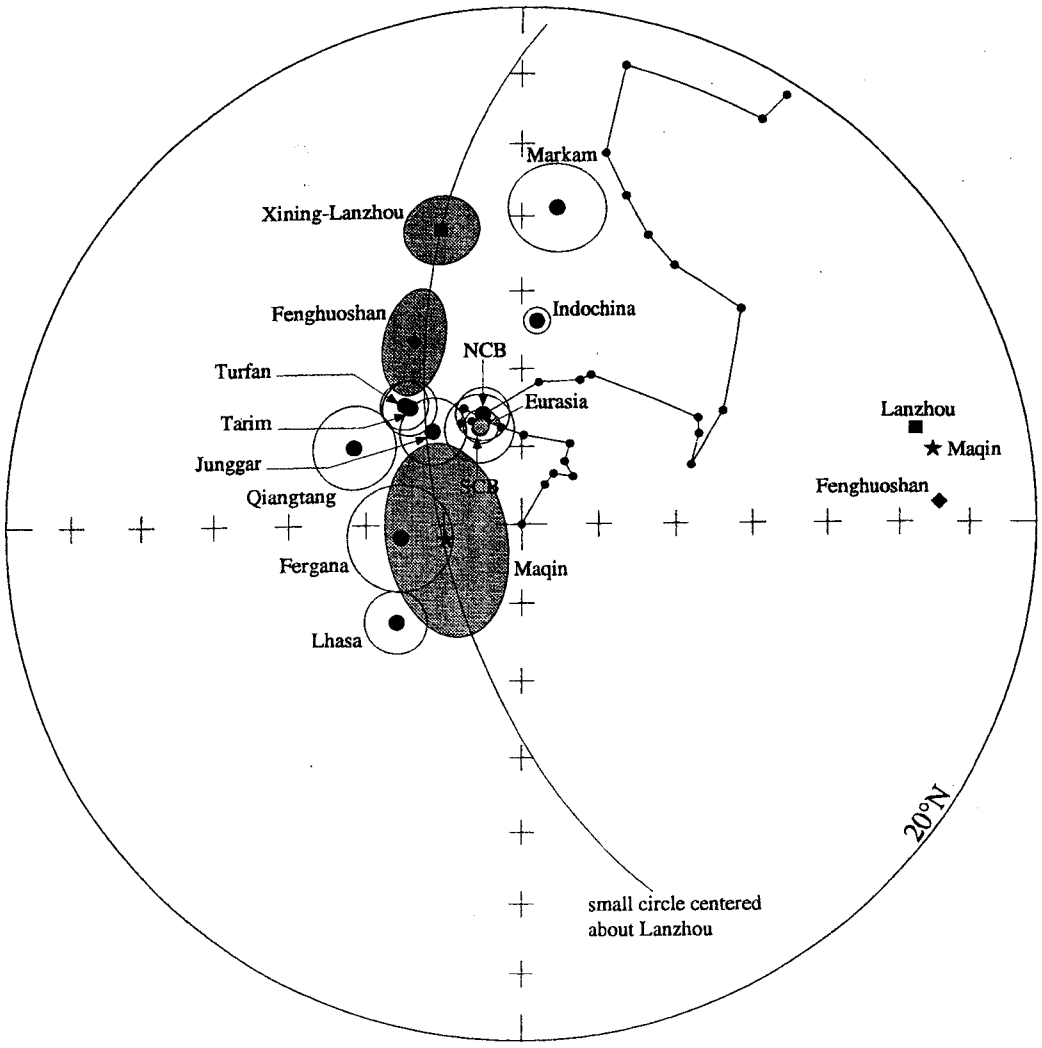


Figure 10. Equal-area projection of Cretaceous poles from the Xining-Lanzhou basin (square), Fenghuoshan-Erdaogou basin (star), and Maqin region (diamond), compared with those of the adjacent Asian blocks and to the reference APWP (small dots, one point every 10 Myr) for Eurasia [after *Besse and Courtillot*, 1991]. References for pole locations are: Eurasia [*Enkin et al.*, 1992], NCB (our compilation), Qiangtang block [*Chen et al.*, 1993a; this study], Lhasa Block [*Pozzi et al.*, 1982; *Achache et al.*, 1984; *Lin and Watts*, 1988; *Chen et al.*, 1993b], Markam [*Otofuji et al.*, 1989; *Huang et al.*, 1992; *Chen et al.*, 1993b], Tarim [*Chen et al.*, 1992; *Li et al.*, 1988], Turfan basin [*Cogné et al.*, 1995], Junggar [*Chen et al.*, 1991], Fergana basin [*Bazhenov*, 1993], SCB [*Lin*, 1984; *Kent et al.*, 1987; *Zhu et al.*, 1988; *Hu et al.*, 1990; *Chan*, 1991; *Huang and Opdyke*, 1992; *Chen et al.*, 1993b], Indochina [*Yang and Besse*, 1993; *Richter et al.*, 1993; *Huang and Opdyke*, 1993; *Funahara et al.*, 1993].

Restoring this motion (Figure 11), the Xining-Lanzhou area would have been located to the NE of a large ENE-WSW trending, combined Tarim-Qaidam block. This block would then have been shortened in the Qilian Shan due to NE propagation of the Tibetan plateau as proposed by *Métivier* [1996]. Differential convergence could have implied slight clockwise rotation of the entire block (Tarim-Qaidam-Qilian Shan) with respect to blocks located farther north, about an axis located in the east. This is consistent with previous analyses of tectonics in the Tian Shan [*Avouac and Tapponnier*, 1992; *Métivier and Gaudemer*, 1997].

The paleoposition of the Kunlun block is weakly constrained: the only available datum is our new result from Maqin area, implying a paleolatitude of 25±6°, consistent with the Tarim-Qaidam assemblage to the north. The actual original shape of this block, which has been severely deformed since the onset of the Indian collision, is difficult to assess. So is its relationship with the Songpan-Garze accretionary complex to the east and its extension to the west. We have tentatively outlined the Kunlun as an elongated zone, whose ends may have been shortened (in a N-S direction), to the west between Qiangtang and Tarim and to the east between Qaidam and the southeast Asian blocks.

Most Cretaceous paleomagnetic results from these southeast Asian blocks display consistent, rather high paleolatitudes: 22.4±1.5° for the Khorat plateau in Indochina [*Yang and Besse*, 1993], 21.2±6.5° (Mengla) to 28.8±8.0° (Xiaguan) for the Simao(Shan-Thai) block [*Huang and Opdyke*, 1993]. The Cretaceous paleolatitude of Southeast Asian blocks are therefore similar to those of Tarim and Qaidam or Kunlun, much to the

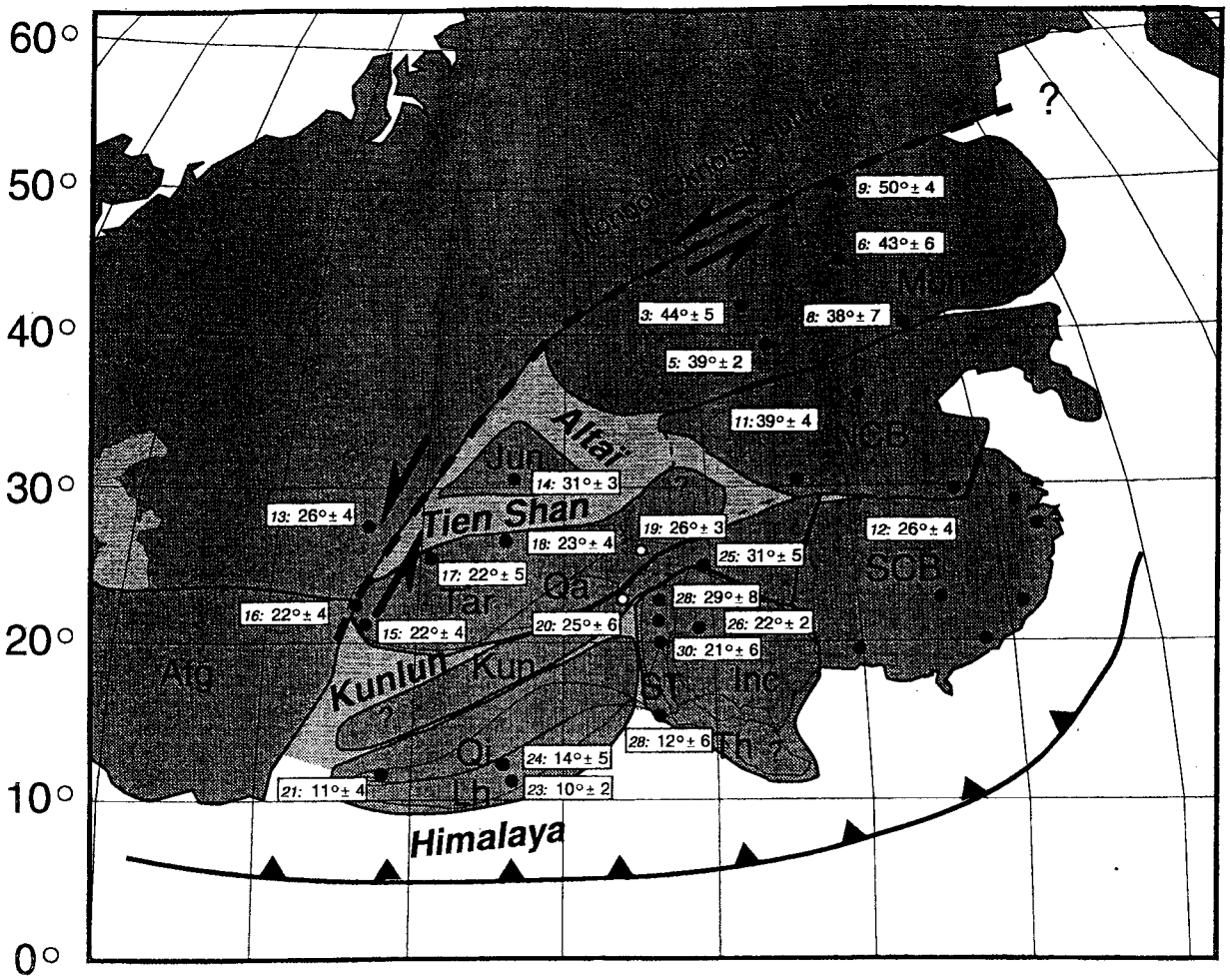


Figure 11. Schematic Cretaceous paleogeographic reconstruction of Asia. Labels indicate the main paleomagnetic sampling locations, their reference number (in italics) as quoted in Table 5, and their Cretaceous paleolatitude. White dots are results from the present study, the sites for Early Tertiary Fenghuoshan results are not shown in this Cretaceous reconstruction. Present-day contours of blocks are drawn as thin lines. See Figure 1 for explanation of abbreviations.

north of the Tibetan blocks (Lhasa and Qiangtang). Only the Kalaw red beds of Myanmar [*Richter et al.*, 1993] provide a lower paleolatitude of 12°±6° which indicates a Late Jurassic-Early Cretaceous paleoposition at the southern border of this assemblage. This is consistent with the idea that these blocks suffered major extrusion to the southeast following the collision [*Tapponnier and Molnar*, 1979; *Peltzer and Tapponnier*, 1988]. *Yang and Besse* [1993] argue for 1500±800 km for the amount of this extrusion, implying an early age for the beginning of deformation. As already suggested by *Yang and Besse* [1993] and *Chen et al.* [1993b], the high latitudes found by *Otofuji et al.* [1989] and *Huang and Opdyke* [1992] for the Markam area contradict a Tibetan block origin. Rather, the Markam area should have been located at the northern boundary of the southeast Asian blocks, to the north of Indochina, or on the southern border of the Kunlun-Songpan-Garze block (Figure 11).

4.3. A Major Discrepancy

The agreement between Cretaceous paleolatitudes for the Qaidam, Tarim, Kunlun, and Junggar argues in favor of an early assemblage of these blocks (the QKTJ assemblage), but these are very different from those expected for Siberia, Mongolia, or northern China. The data from *Pruner* [1988] and our own unpublished data from the Mongol-Okhotsk suture near 125°E longitude argue for lack of significant post-Cretaceous convergence between Mongolia and Siberia. Results of *Zhao et al.* [1990], *Chen et al.* [1993b], *Ma et al.* [1993], and *Gilder et al.* [1996] argue in a similar way for the same post Cretaceous motion between Siberia and the North China Block (see the average pole calculated above, Table 5 and Figure 10). It is therefore proposed that Siberia, Mongolia, and north China were close to their present respective positions in the Cretaceous (Figure 11), at least as far as N-S motions are concerned.

This is in striking contrast with the apparently large northward movement of the Lanzhou-Xining or Kunlun (Maqin) areas with respect to Siberia implied by our new data (770±500 km and 660±700 km, respectively). There appears to have been very significant convergence of the QKTJ assemblage and the main part of Eurasia located to the north and east. Up to 700 km of the 2700 km of India-Eurasia convergence following the collision would have been taken up to the north of the Junggar, Tarim and Qaidam. This strengthens the puzzling problem

already pointed out in several earlier paleomagnetic studies of central Asian blocks [*Chen et al.*, 1993b; *Frost et al.*, 1995; *Gilder et al.*, 1996].

Yet the mountain ranges to the north do not seem to be either high and/or wide enough to account for this amount of N-S convergence solely by crustal shortening. *Avouac* [1991] estimates that only some 200 km at the most could have been absorbed by that mode of intraplate shortening. This leaves us with some 500 km of north-south motion to be accounted for in another way (granted with a large 95% uncertainty of the order of 500 km).

We see in the tentative Cretaceous reconstruction of Figure 11 that convergence in the Altai may have taken place together with left-lateral strike-slip motion of the QKTJ blocks with respect to Siberia along their joint NW boundary. Evidence for this large fault zone is lacking, and its actual paleolocation is not easily identified. It might be tempting to place it along a line joining the Pamir ranges, at the NW corner of the Indian indentor, to the lake Baikal region. However, this appears to be in contradiction with paleomagnetic results from the Fergana basin [*Bazhenov*, 1993; *Thomas et al.*, 1994] which display paleolatitudes in agreement with those from the Tarim block. Should our hypothesis be valid, a location of the main strike-slip fault system farther to the west is required.

Another consequence of this interpretation is that the left-lateral strike-slip fault should continue to the NE, implying shear motion along, for instance, the Mongol-Okhotsk suture and the eastward extrusion of the Mongolia-NCB-SCB assemblage. Such extrusion is consistent in geometry with tectonic interpretations of *Tapponnier and Molnar* [1979] (see also *Molnar and Tapponnier* [1975]), however with much larger amplitudes. A growing paleomagnetic database assembled over the last decade and complemented in this paper indicates major problems in the relative Cretaceous and Tertiary paleolatitudes of Asian blocks. Although the possibilities of dating errors, unremoved secondary overprints and nondipole field geometry have all been advocated [*Zhao et al.*, 1990; *Chen et al.*, 1993b; *Ma et al.*, 1993; *Frost et al.*, 1995; *Gilder et al.*, 1996; *Bazhenov and Burtman*, 1986; *Thomas et al.*, 1993; *Chauvin et al.*, 1996], the Cretaceous database is now such that it is hard to escape the alternate possibility that, as has already happened in studies of other remote areas, paleomagnetism points to major previously undetected tectonic motions in central Asia. The role of a number of ranges in absorbing part of the shortening (Qilian Shan, Tian Shan, Altai, etc.) remains to be further assessed, and geological work should focus on possible identification of major intracontinental deformation zones such as the Mongol-Okhotsk suture and its elusive continuation to the SW, as suggested here. After close to 20 years of paleomagnetic work in Asia, much remains to be done in order to better resolve a picture of deformation of the continental mosaic following the impact of India.

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